

Online Mobile Shop

A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

**DARSHAN INSTITUTE OF ENGINEERING AND TECHNOLOGY,
RAJKOT**



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March – 2026



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CERTIFICATE

This is to certify that the project report submitted along with the project entitled “ONLINE MOBILE SHOP” has been carried out by “**Vekariya Parth**” under my guidance in partial fulfilment for the degree of Bachelor of Engineering in **Department of Computer Science and Engineering**, 8th Semester of Darshan University, Rajkot during the academic year 2025-26.

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DECLARATION

We hereby declare that the Project report submitted along with the Project entitled “ONLINE MOBILE SHOP” submitted in partial fulfilment for the degree of Bachelor of Technology in **Computer Science and Engineering** to Darshan University, Rajkot, is a bonafide record of original project work carried out by me at Darshan Institute of Engineering and Technology under the supervision of Prof. Kalpesh Surati and that no part of this report has been directly copied from any students’ reports or taken from any other source, without providing due reference.

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ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my project guide Prof. Kalpesh Surati and all the faculty members for helping me through my project by giving me the necessary suggestions and advices along with their valuable co-ordination in completing this work.

I also thank my parents, friends and all the members of the family for their precious support and encouragement which they had provided in completion of my work. In addition to that, I would also like to mention the college personals who gave me the permission to use and experience the valuable resources required for the project from the college premises.

Thus, in conclusion to the above said, I once again thank the faculties and members of **Darshan Institute of Engineering & Technology** for their valuable support in completion of the project.

With Sincere Regards

Vekariya Parth

ABSTRACT

Online Mobile Shop

The **Online Mobile Shop** is a web-based application designed to provide an easy and convenient platform for customers to purchase mobile phones online. The system allows users to browse various mobile brands, models, storage options, and prices from a single website. Customers can register, log in, and place orders for their preferred mobile phones while also managing their delivery addresses and viewing their order history. The application includes an admin panel that enables the administrator to manage mobile details, brands, storage variants, and customer orders. The admin can update the order status from pending to success, which is then reflected in the user's order history. The system is developed using React for the frontend, ASP.NET Web API for the backend, and SQL Server for database management. The main objective of this project is to simplify the mobile purchasing process, reduce manual work, and provide a fast, secure, and user-friendly online shopping experience.

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Traditional mobile shops mainly rely on manual systems for managing product records, billing, and customer information. This approach is time-consuming and increases the chances of calculation errors and data mismanagement. Customers also face difficulty in checking product availability, comparing specifications, and tracking their orders. In the modern digital world, users expect fast, secure, and convenient online services.

The motivation behind developing the **Online Mobile Shop** system is to create a web-based application that automates inventory management, user registration, and order processing. The system allows customers to browse mobile phones, view storage variants, and place orders easily from anywhere. It also helps the admin manage brands, products, and bill status efficiently. This project improves accuracy, reduces manual workload, and enhances the overall customer experience.

1.2 Project Scope

The Online Mobile Shop system is designed to provide a web-based platform for managing mobile products and online orders. It allows customers to browse mobile phones, check specifications, select storage variants, and place orders securely. The admin can manage brands, phone details, storage options, users, and bill status through an admin dashboard. The system ensures secure authentication using JWT and maintains proper order history for users. This project improves inventory control, billing efficiency, and overall customer experience.

1.3 Project Stakeholders

The project stakeholders for the **Online Mobile Shop** system include:

- **Admin**– Manages the entire system including adding/updating brands, phone details, storage variants, and updating bill status from pending to success.
- **Software Developers** – Develop and maintain the frontend (React) and backend (.NET API) of the system.
- **Database Administrator** – Handles database management, relationships between tables, and data security.
- **Customers**– Register into the system, browse mobile phones, place orders, and view their bill history.

1.4Project Scheduling

Phase	Tasks	Duration
Week 1-2	Requirement Analysis & Feasibility Study	2 Weeks
Week 3	System Design (ERD, DFD, UML Diagrams)	1 Week
Week 4-6	Database Design & Setup (SQL Server)	3 Weeks
Week 7-10	Frontend Development (React, HTML, CSS, JavaScript)	4 Weeks
Week 11-14	Backend Development (ASP.NET Web API with JWT)	4 Weeks
Week 15-16	Integration & Testing	2 Weeks
Week 17	Deployment & Documentation	1 Week
Week 18	Final Review & Submission	1 Week

CHAPTER 2

SYSTEM ANALYSIS

2.1 Introduction

The Online Mobile Shop system is developed to convert traditional mobile shop operations into a web-based application. It allows customers to browse mobile phones, check specifications, and place orders online. The system also provides an admin panel to manage brands, products, storage variants, and bill status. This application improves efficiency, accuracy, and overall business management.

2.2 Requirement Analysis

2.2.1 Functional Requirement

Admin:

- Add, update, and delete mobile brands.
- Add, edit, and manage phone details and storage variants.
- View registered users and manage their data.
- Update bill status from pending to success.
- View order and sales reports.

User (Customer):

- Register and login securely into the system.
- Browse and search mobile phones.
- View phone specifications and storage options.
- Add products to cart and place orders.
- View bill history and order status.

2.2.2 Non-Functional Requirement

The non-functional requirements define the quality attributes of the **Online Mobile Shop** system .

Performance:

- The system should handle multiple users simultaneously without performance issues.
- Product search and page loading time should be less than 2–3 seconds. Order processing should be fast and efficient.

Security:

- User authentication must be implemented using JWT token-based authentication.
- Passwords should be encrypted and stored securely in the database.
- Only authorized admin users can manage products and update bill status.

Scalability:

- The system should support the addition of new brands, products, and users without major system changes.
- The database should efficiently handle a large number of product and order records.

Usability:

- The user interface should be simple, responsive, and easy to navigate on desktop and mobile devices.

2.2.3 Software & Hardware Requirements

User	Particulars	Client System	Server System
Admin	O/S	Windows 10 / 11	Windows Server
	Processor	Dual Core (Minimum)	Intel i5 / Higher
	Hard Disk	100 GB (Minimum)	500 GB (Minimum)
	RAM	4 GB (Minimum)	8 GB (Minimum)
	Software	Web Browser ,Node.js	.NET ,SQL Server
Customer	Operating System	Windows / Android / iOS	—
	Processor	Dual Core (Minimum)	—
	Internal Memory	2 GB (Minimum)	—
	RAM	2 GB (Minimum)	—
	Software	Web Browser	—

2.3 Design and Implementation Constraints

The Online Mobile Shop system is developed as a web-based application using React for frontend and ASP.NET Web API for backend. The system requires a stable internet connection for proper communication between client and server. JWT authentication is implemented for secure access, so proper token management is necessary. The database design must maintain proper relationships between tables such as PhoneDetail, PhoneBrand, User, and Bill. Only the admin has permission to manage products and update order status, ensuring controlled access and data security.

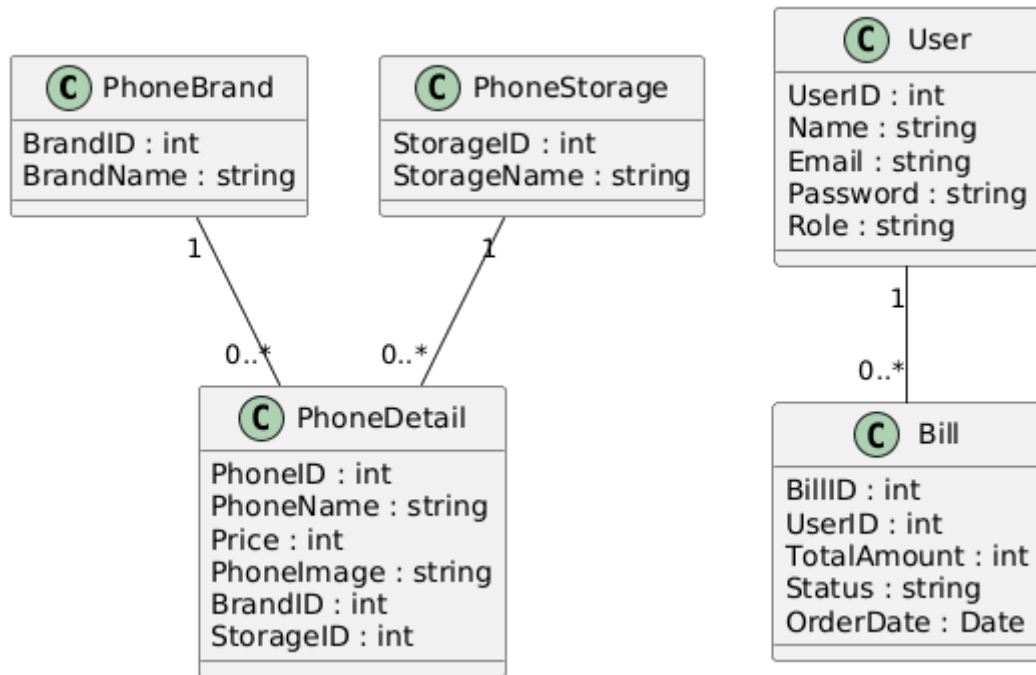
2.4 Assumptions and Dependencies

- Internet connection is required for communication between client and server.
- The system must be hosted on a server that supports ASP.NET Web API and SQL Server.
- Admin must have valid login credentials to manage brands, phone details, storage, and bill status.
- Each user must have a unique ID and registered account to place orders and view bill history.

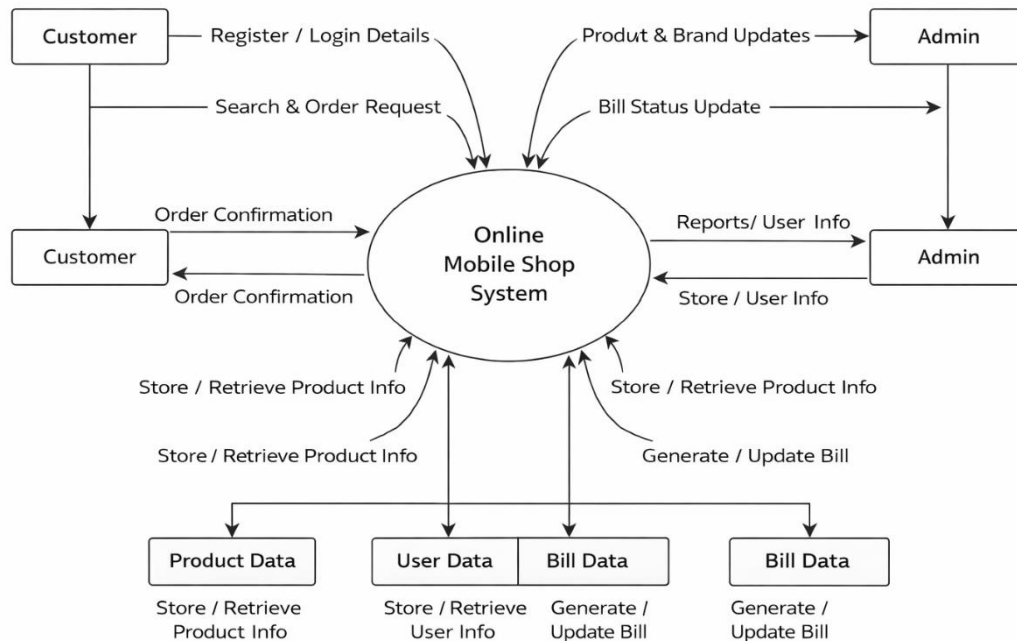
CHAPTER 3

SYSTEM DESIGN

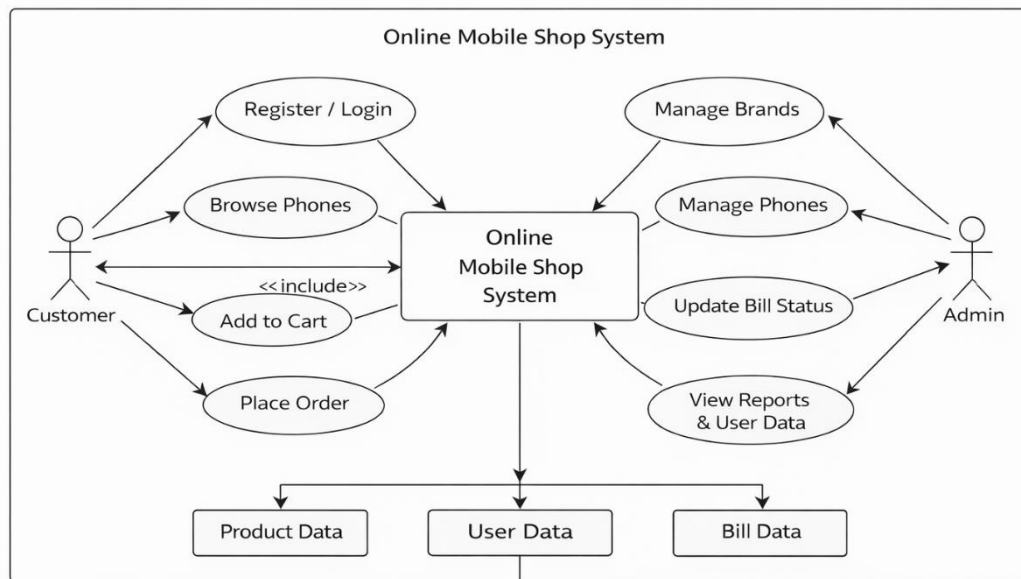
3.1 UML Diagram



3.1.1 Data-Flow Diagram



3.1.2 Use-case Diagram



3.1.3 Use-case Scenarios

1.Search Article

Abbreviated Title	Login
Actors	Admin, Customer
Description	To interact with the Online Mobile Shop system, the user must log in with valid credentials. The system validates the registration details and allows access according to the user role.
Pre-Conditions	User must be registered in the system.
Task Sequence	• System shows Login Screen • User enters username and password • User clicks Login button • System validates credentials
Post Conditions	System transfers control to respective dashboard (Admin Panel or Customer Home Page).
Exception	If credentials are invalid, system displays “Invalid Username or Password” message and redirects to Login screen.

2.Search Mobile

Abbreviated Title	Search Mobile
Actors	Customer
Description	Search mobile allows customers to browse and search available mobile phones using brand name, model, or price range.
Pre-Conditions	User must be logged in.
Task Sequence	• System displays search page • User enters search criteria • User clicks search button • System displays matching mobile phones
Post Conditions	User can view product details and proceed to add to cart.
Exception	If no product matches search, system displays “No products found”.

3.Add to Card

Abbreviated Title	Add to Cart
Actors	Customer
Description	Customer can add selected mobile phones to the shopping cart before placing an order.
Pre-Conditions	User must be logged in and product must be available.
Task Sequence	• User selects mobile phone • User clicks Add to Cart • System stores product in cart
Post Conditions	Selected product is added to user cart successfully.
Exception	If product is out of stock, system displays availability message.

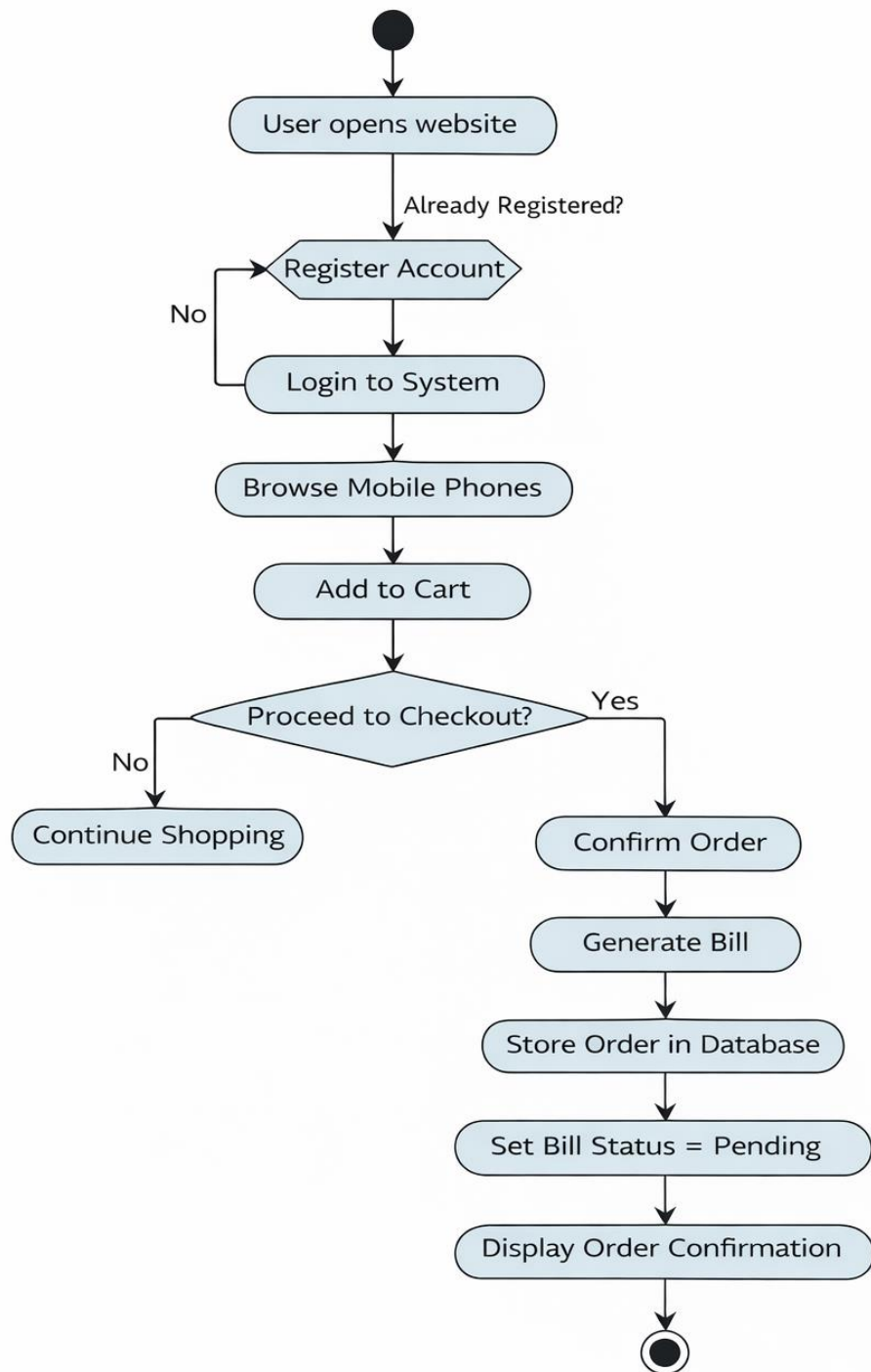
4.Place Order

Abbreviated Title	Place Order
Actors	Customer
Description	Customer confirms the cart items and places an order. The system generates bill and stores order details.
Pre-Conditions	User must be logged in and cart must contain items.
Task Sequence	• User reviews cart • User confirms order • System generates bill • System stores order details
Post Conditions	Order is successfully placed and bill status is set to pending.
Exception	If payment or order process fails, system displays error message.

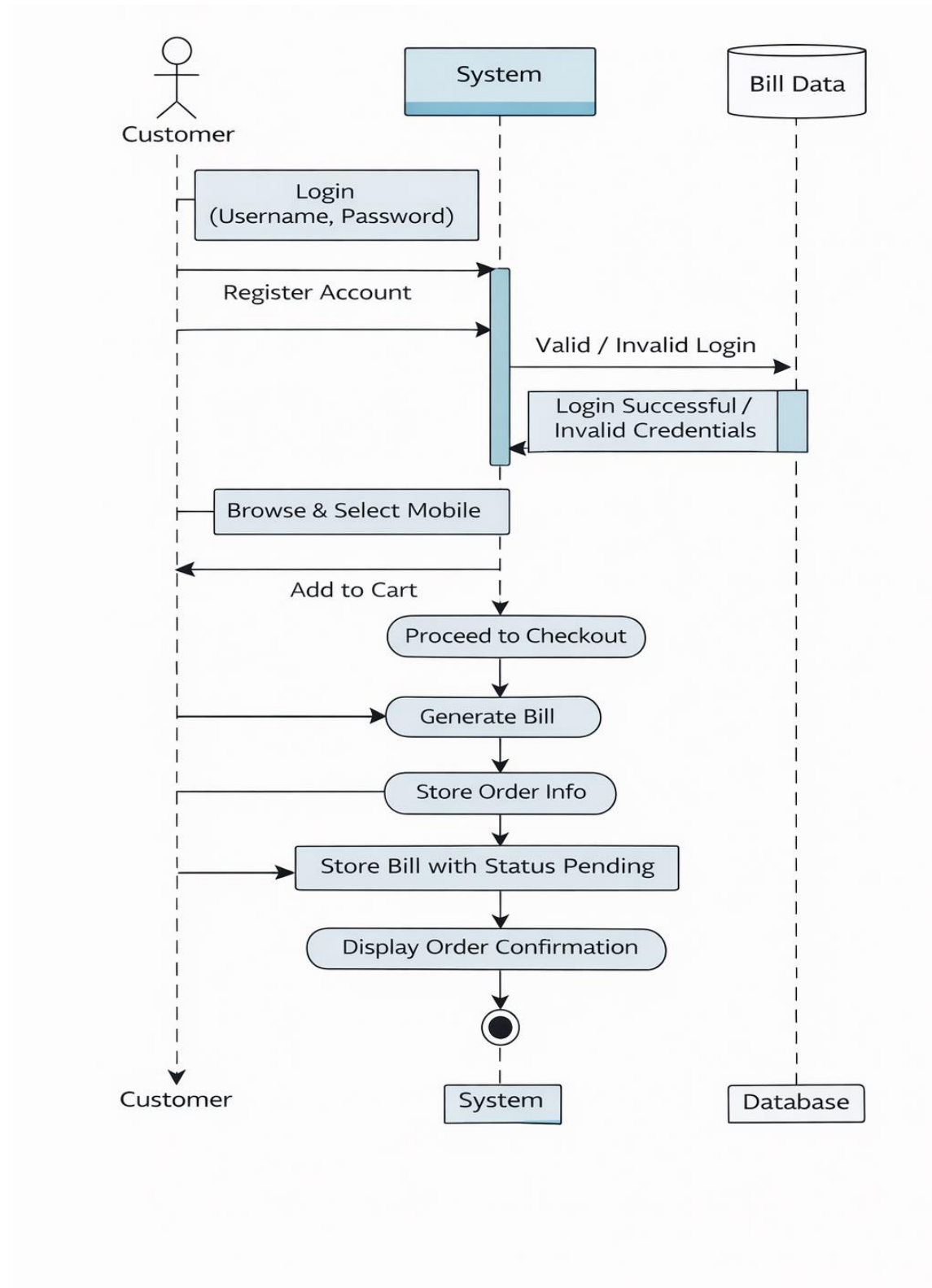
5.Manage Products

Abbreviated Title	Manage Products
Actors	Admin
Description	Admin can add, update, or delete mobile phones and manage brand and storage information.
Pre-Conditions	Admin must be logged in.
Task Sequence	• System displays product management panel • Admin selects add/update/delete option • System saves changes in database
Post Conditions	Product information is updated successfully.
Exception	If invalid data is entered, system displays validation error.

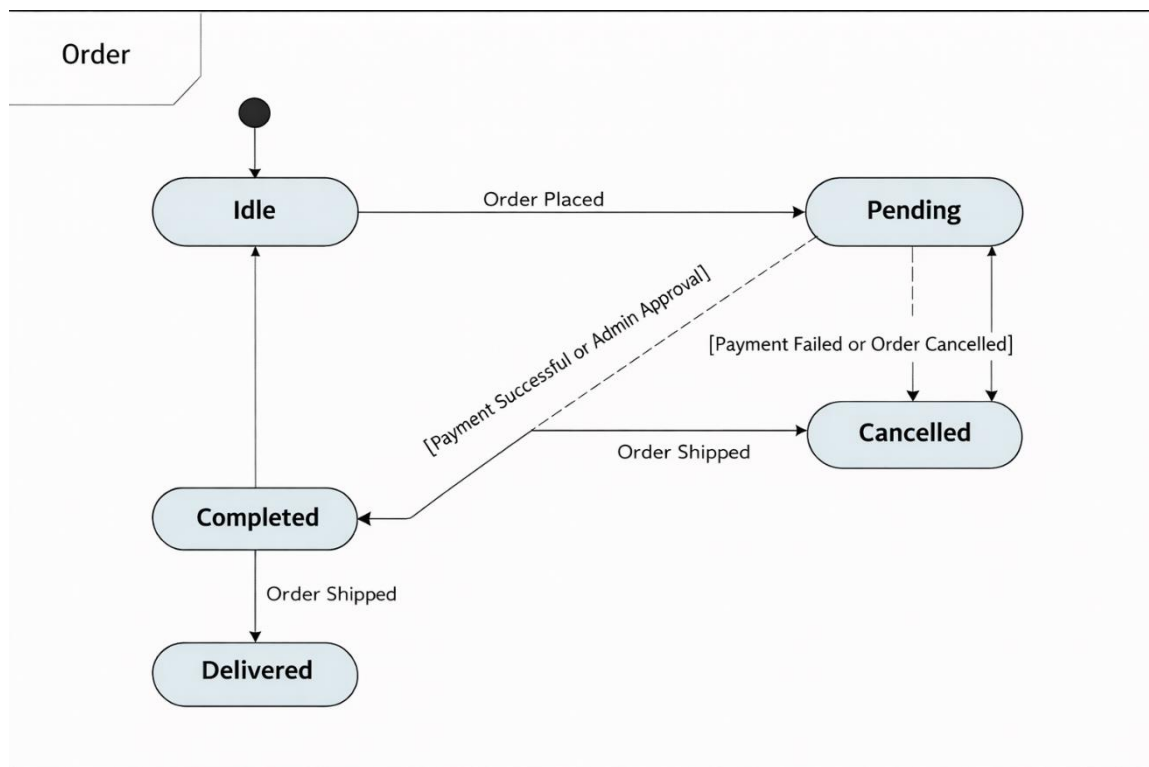
3.1.4 Activity Diagram



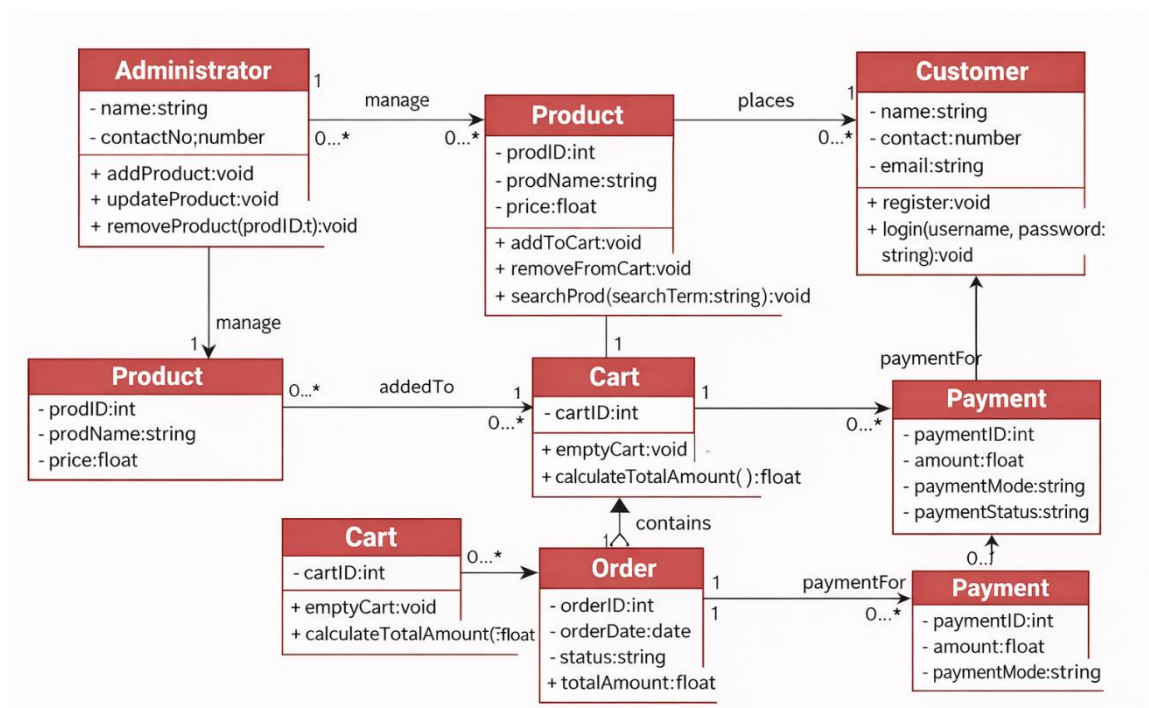
3.1.5 Sequence Diagram



3.1.6 State Diagram



3.1.7 Class Diagram



3.2 Database Design

3.2.1 Database Schema

Table 1: PhoneBrand

Column	Data Type	Null	Keys & Constrains	Default Value & Description
BrandID	int	NN	PK (Auto Increment)	Unique Brand ID
BrandName	varchar(100)	NN	—	Name of Mobile Brand

Table 2: PhoneStorage

Column	Data Type	Null	Keys & Constrains	Default Value & Description
StorageID	int	NN	PK (Auto Increment)	Unique Storage ID
StorageName	varchar(50)	NN	—	Storage Variant (e.g., 128GB, 256GB)

Table 3: PhoneDetail

Column	Data Type	Null	Keys & Constrains	Default Value & Description
PhoneID	int	NN	PK (Auto Increment)	Unique Phone ID
PhoneName	varchar(150)	NN	—	Name of Mobile Phone
Price	int	NN	—	Price of Mobile
PhoneImage	varchar(255)	AN	—	Image Path of Phone
BrandID	int	NN	FK (PhoneBrand)	Reference to Brand Table
StorageID	int	NN	FK (PhoneStorage)	Reference to Storage Table

Table 4: User

Column	Data Type	Null	Keys & Constrains	Default Value & Description
UserID	int	NN	PK (Auto Increment)	Unique User ID
Name	varchar(100)	NN	—	User Full Name
Email	varchar(150)	NN	Unique	User Email Address
Password	varchar(255)	NN	—	Encrypted Password
Role	varchar(50)	NN	—	User Role (Admin/Customer)

Table 5: Address

Column	Data Type	Null	Keys & Constrains	Default Value & Description
AddressID	int	NN	PK (AutoIncrement)	Unique Address ID
UserID	int	NN	FK (User)	Reference to User Table
City	varchar(100)	NN	—	City Name
State	varchar(100)	NN	—	State Name
Pincode	varchar(10)	NN	—	Postal Code

Table 6: Bill

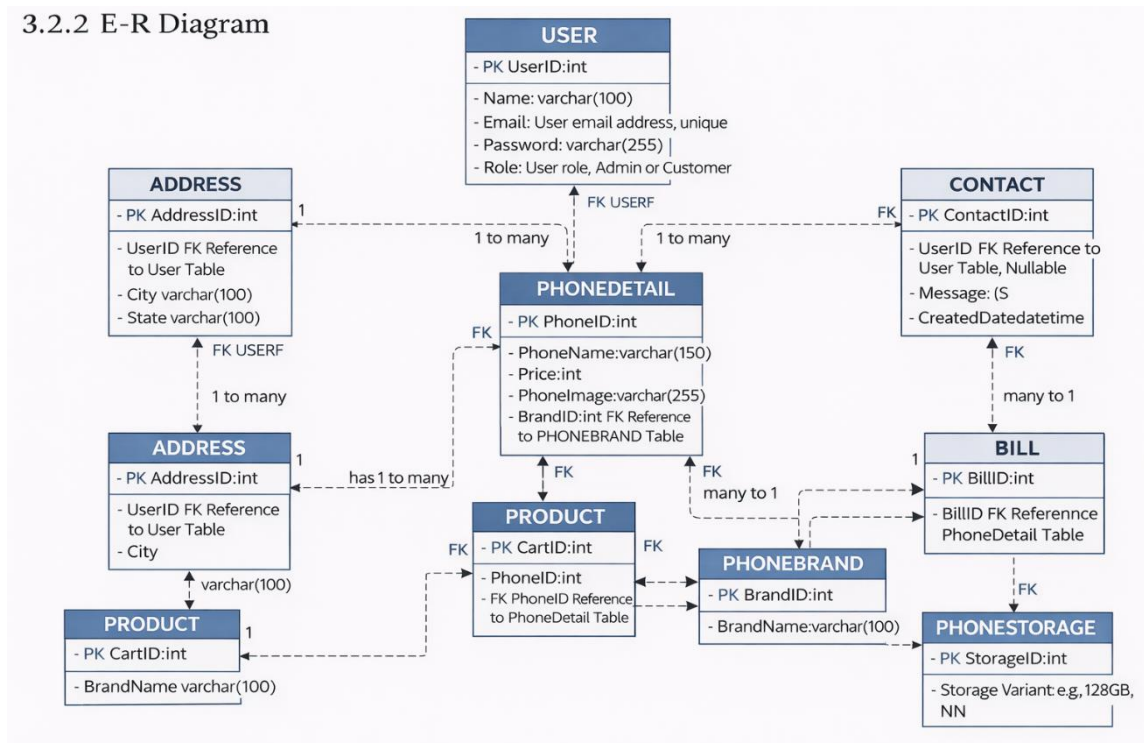
Column	Data Type	Null	Keys & Constrains	Default Value & Description
BillID	int	NN	PK (Auto Increment)	Unique Bill ID
UserID	int	NN	FK (User)	Reference to User Table
TotalAmount	int	NN	—	Total Bill Amount
Status	varchar(50)	NN	Default: 'Pending'	Order Status (Pending/Success)
OrderDate	datetime	NN	—	Date of Order

Table 7: Contact

Column	Data Type	Null	Keys & Constrains	Default Value & Description
ContactID	int	NN	PK (AutoIncrement)	Unique Contact ID
UserID	int	AN	FK (User)	Reference to User Table
Message	varchar(500)	NN	—	Customer Message
CreatedDate	datetime	NN	—	Message Date

3.2.2 E-R Diagram

3.2.2 E-R Diagram

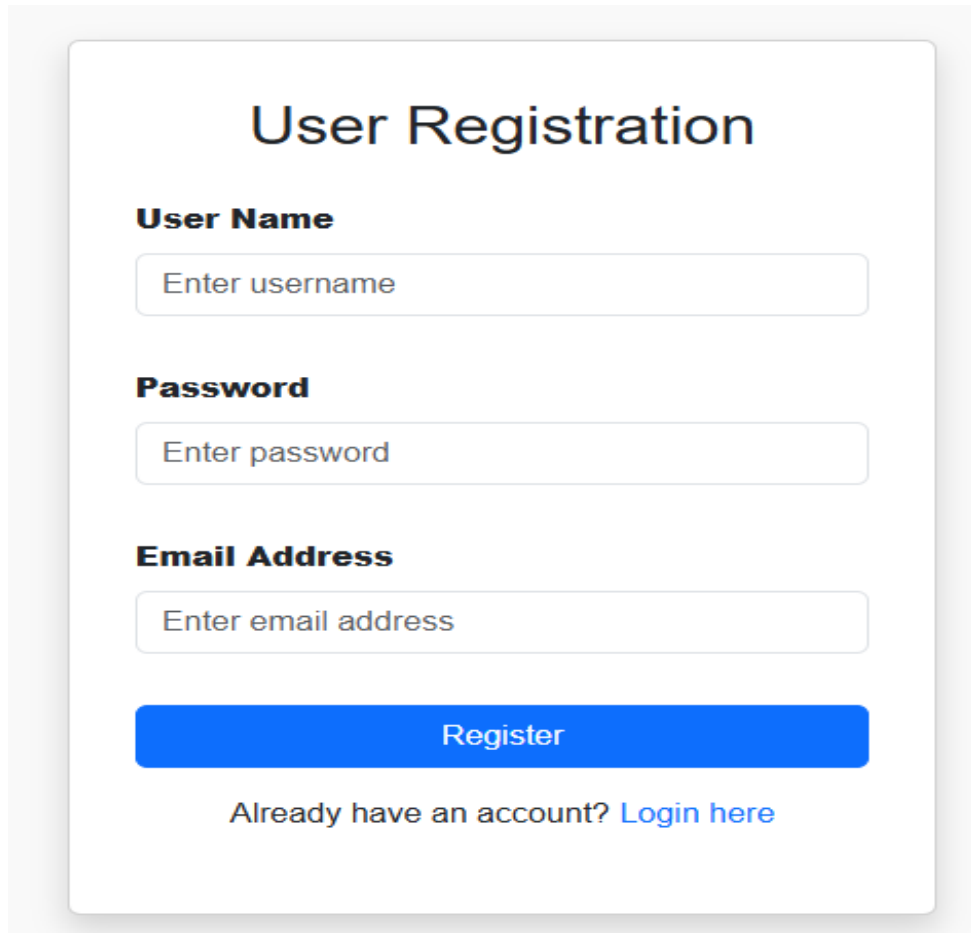


CHAPTER 4

IMPLEMENTATION

4.1 Module Design and Specification

4.1.1 Screen-1: Registration Form

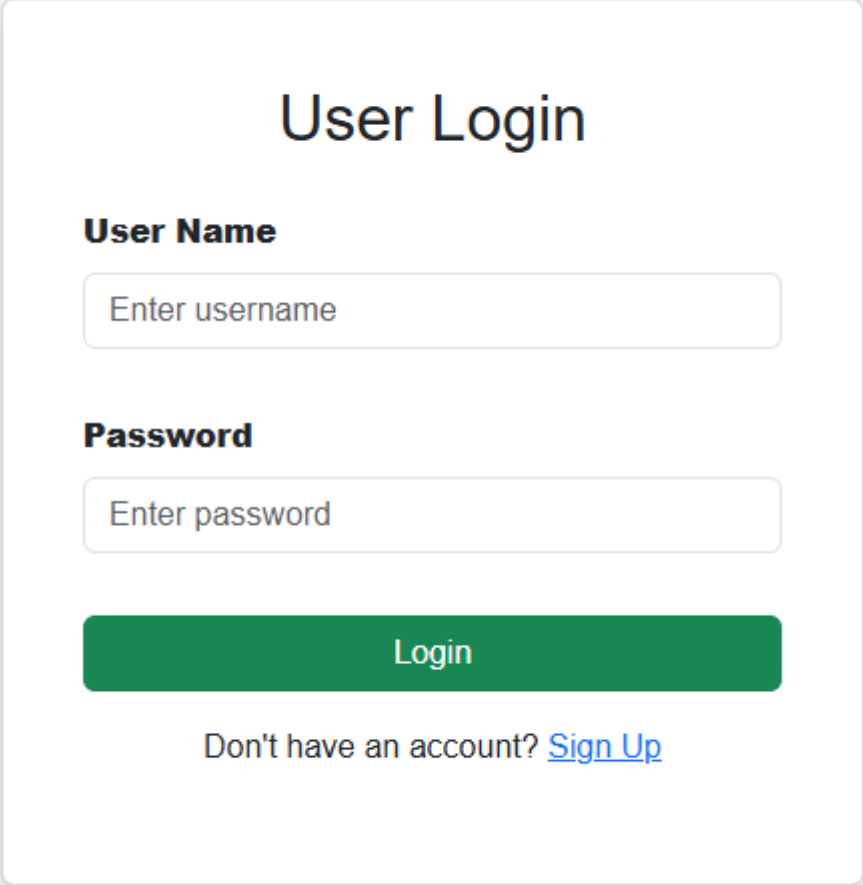


The image shows a 'User Registration' form. It has a title 'User Registration' at the top. Below the title are three input fields: 'User Name' with a placeholder 'Enter username', 'Password' with a placeholder 'Enter password', and 'Email Address' with a placeholder 'Enter email address'. Below these fields is a blue 'Register' button. At the bottom, there is a link that says 'Already have an account? Login here'.

Purpose: This form allows new users to register in the Online Mobile Shop system.

Sr.	Screen Element	Input Type	O/M	Description
1	User Name	Textbox	M	Editable field to enter username.
2	Email Address	Textbox	M	Accepts valid email format.
3	Password	Password	M	Accepts password and displays as dots/stars.
4	Register	Button	—	Stores entered data into database after validation.
5	Login here	Link	—	Redirects registered users to Login page.

4.1.2 Screen-2: Login Form

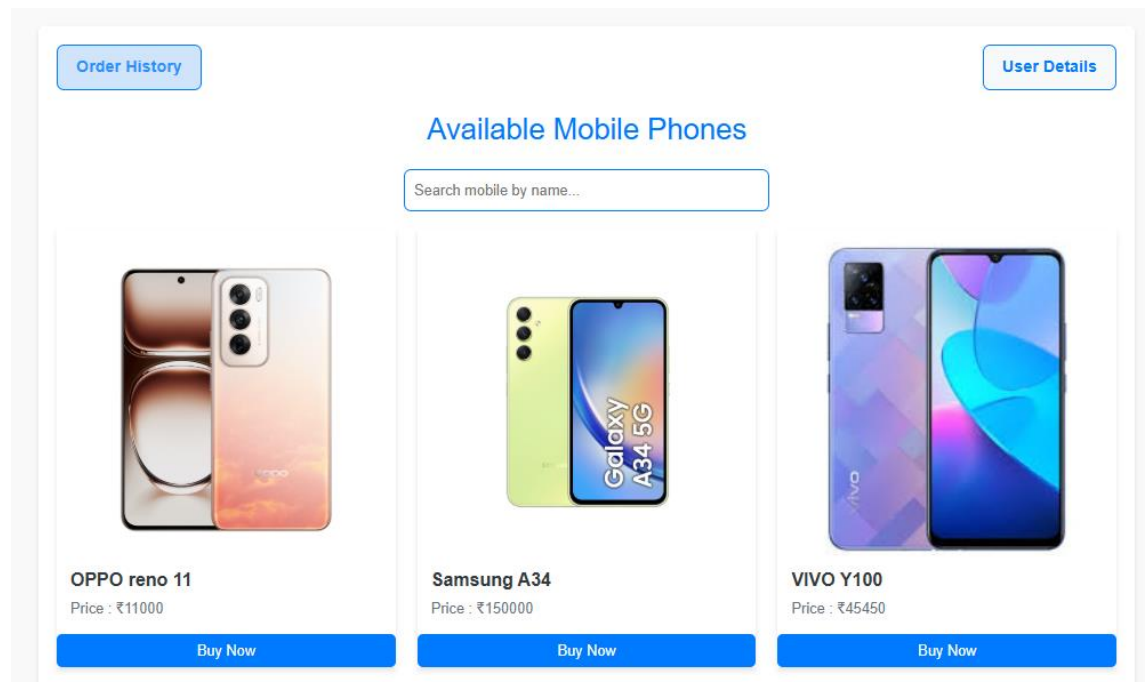


The image shows a 'User Login' form. It has a title 'User Login' at the top. Below the title are two input fields: 'User Name' with a placeholder 'Enter username' and 'Password' with a placeholder 'Enter password'. Below these fields is a green 'Login' button. At the bottom, there is a link that says 'Don't have an account? [Sign Up](#)'.

Purpose: This form allows registered users to log into the system.

Sr.	Screen Element	Input Type	O/M	Description
1	User Name	Textbox	M	Accepts registered username.
2	Password	Password	M	Accepts password and hides characters.
3	Login	Button	—	Validates credentials and redirects to home page.
4	Sign Up	Link	—	Redirects new users to registration page.

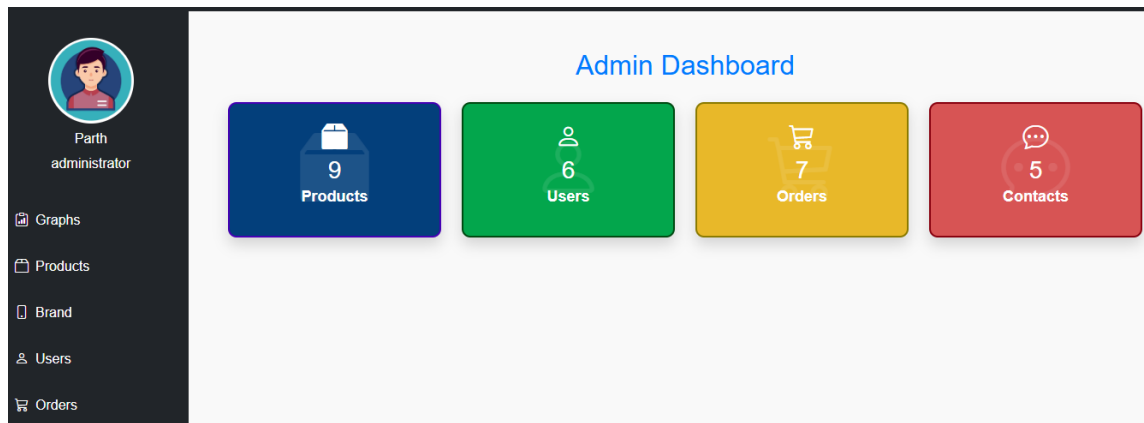
4.1.3 Screen-3: Product Listing Page



Purpose: This screen displays all available mobile phones for customers.

Sr.	Screen Element	Input Type	O/M	Description
1	Search Bar	Textbox	O	Allows user to search mobile by name.
2	Product Image	Image	—	Displays mobile phone image.
3	Product Name	Label	—	Displays mobile phone name.
4	Price	Label	—	Displays mobile price.
5	Buy Now	Button	—	Redirects user to purchase or cart page.
6	Order History	Button	—	Redirects user to order history page.
7	User Details	Button	—	Displays user profile details.

4.1.4 Screen-4: Admin Dashboard



Purpose: This screen displays all available mobile phones for customers

Sr.	Screen Element	Input Type	O/M	Description
1	Products Card	Display Panel	—	Shows total number of products.
2	Users Card	Display Panel	—	Shows total number of users.
3	Orders Card	Display Panel	—	Shows total number of orders.
4	Contacts Card	Display Panel	—	Shows total number of contact messages.
5	Sidebar Menu	Navigation	—	Contains options: Graphs, Products, Brand, Users, Orders.

4.2 Testing

4.2.1 Functional Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC001	User Login	1. Open Login Page 2. Enter valid username & password 3. Click "Login"	User is successfully logged in and redirected to Product Listing page	Pass
TC002	Invalid Login	1. Enter incorrect credentials 2. Click "Login"	Error message displayed: "Invalid Username or Password"	Pass
TC003	User Registration	1. Open Registration page 2. Enter valid details 3. Click "Register"	User account is successfully created	Pass
TC004	Search Mobile	1. Enter mobile name in search bar 2. Click search	Matching mobile phones are displayed	Pass
TC005	Buy Product	1. User selects mobile 2. Click "Buy Now" 3. Confirm order	Order is placed and bill is generated with status Pending	Pass
TC006	Add New Product (Admin)	1. Admin logs in 2. Navigate to "Products" 3. Enter product details 4. Click "Save"	Product is successfully added to database	Pass
TC007	Update Bill Status	1. Admin logs in 2. Navigate to "Orders" 3. Update status from Pending to Success	Bill status is successfully updated	Pass
TC008	Logout	1. Click on "Logout" button	User is logged out and redirected to login page	Pass

4.2.2 Non-Functional Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
NFT001	Performance Testing	1. Open Product Listing page 2. Load multiple products	Page loads within acceptable time (2–3 seconds)	Pass
NFT002	Security Testing	1. Try accessing Admin Dashboard without login	System redirects to Login page	Pass
NFT003	JWT Authentication	1. Login as user 2. Access protected API endpoint without token	Access denied without valid token	Pass
NFT004	Usability Testing	1. Navigate through registration, login, and product pages	Interface is user-friendly and easy to navigate	Pass
NFT005	Compatibility Testing	1. Open system in Chrome, Edge, and Firefox	System works properly in all major browsers	Pass
NFT006	Database Integrity	1. Add product 2. Verify data in database	Data is stored correctly without duplication	Pass
NFT007	Error Handling	1. Submit empty form fields	Proper validation messages are displayed	Pass

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

The Online Mobile Shop system has been successfully designed and implemented as a web-based application using React for the frontend and ASP.NET Web API for the backend. The system provides an efficient platform for customers to browse mobile phones, search products, and place orders online. It also includes a secure authentication mechanism using JWT to ensure safe access to the system.

The admin panel enables effective management of brands, phone details, users, and order status. The system reduces manual work, improves data accuracy, and enhances overall customer experience. Through proper database design and testing, the application ensures reliability, performance, and security.

Thus, the Online Mobile Shop system fulfills all the functional and non-functional requirements and provides a complete digital solution for managing mobile shop operations.

5.2 Future Enhancement

While the current system covers the fundamental aspects of online mobile shop management, several enhancements can be made to improve its functionality and user experience:

1. Mobile Application Integration

Developing an Android/iOS mobile application to allow customers to browse products, place orders, track delivery status, and receive notifications about offers and discounts.

2. Online Payment Gateway Integration

Integrating secure online payment gateways such as UPI, credit/debit cards, and net banking to allow customers to complete transactions directly through the system.

3. Advanced Recommendation System

Implementing an AI-based recommendation system that suggests mobile phones based on customer preferences, search history, and purchase behavior.

4. Real-Time Order Tracking

Enhancing the system to provide real-time order tracking and delivery updates for customers.

APPENDICES

Appendix A: Acronyms and Abbreviations

Acronym	Full Form
OMS	Online Mobile Shop
DBMS	Database Management System
UI	User Interface
API	Application Programming Interface
JWT	JSON Web Token
SQL	Structured Query Language
ER Diagram	Entity Relationship Diagram
UML	Unified Modelling Language
HTTP	HyperText Transfer Protocol
CRUD	Create, Read, Update, Delete

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